

DS	Session	Graph	Question Information	Level 1	Challenge 88
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Question description

You are given a tree consisting of n nodes. A *matching* is a set of edges where each node is an endpoint of at most one edge. What is the maximum number of edges in a matching?

Input Format

The first input line contains an integer n : the number of nodes. The nodes are numbered $1, 2, \dots, n$.

Then there are $n-1$ lines describing the edges. Each line contains two integers a and b : there is an edge between nodes a and b .

Output Format

Print one integer: the maximum number of pairs.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq a, b \leq n$

Example

Input:

5
1 2
1 3
3 4
3 5

Output:

2

Explanation: One possible matching is (1,2) and (3,4)

Logical Test Cases

Test Case 1

INPUT (STDIN)

5
1 2
1 3
3 4
3 5

EXPECTED OUTPUT

2

Test Case 2

INPUT (STDIN)

5
1 1
1 2
2 3
3 4

EXPECTED OUTPUT

6

Mandatory Test Cases

Test Case 1

KEYWORD

void dfs(int p,int i)

Test Case 2

KEYWORD

void link(int i,int j)

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

9

Test Case 2

TOKEN COUNT

350

Test Case 3

NLOC

50